Integration of Stress Testing with Graph Theory to Assess the Resilience of Urban Road Networks: A Case Study of Tianhe District, Guangzhou

Keywords: Urban Resilience; Stress Testing; Graph Theory; Betweenness Centrality; Spatial Heterogeneity

1. Introduction

1.1 Research Background

Vulnerability and Criticality of Urban Transportation Systems

Transportation networks are not only the physical backbone of urban operations but also critical service carriers sustaining socio-economic activities. However, with accelerated global climate change and urbanization, urban transport infrastructure faces increasingly complex risks.

The United Nations Office for Disaster Risk Reduction (*UNISDR*) emphasizes that infrastructure must maintain basic structure and function under disaster disruptions. Facing "acute and chronic stresses" such as earthquakes, extreme rainfall, or sudden public events, the high interconnectivity of urban road networks means that local node failures often lead to systemic paralysis through cascading effects.

Paradigm Shift: From "Capacity" to "Resilience"

Traditional urban planning often focuses on road traffic capacity and supply-demand balance, which is insufficient for addressing highly uncertain external shocks. Resilience implies the ability of a system to "resist, absorb, adapt to, and recover from" disturbances. For a high-density, high-flow core urban area like Tianhe District, relying solely on risk assessment based on known probabilities is inadequate to handle unknown potential threats. Therefore, it is urgent to introduce a resilience perspective and use quantitative methods to assess the topological stability and functional maintenance level of the road network under extreme conditions.

1.2 Research Significance

1.Theoretical Innovation: Integration of Graph Theory and Stress Testing

This study transcends the limitations of traditional transport planning that focuses solely on physical attributes by integrating Complex Network Theory with Stress Testing methodologies. By introducing graph theory metrics such as Betweenness Centrality and Network Efficiency, we construct an analytical framework capable of quantitatively assessing the performance degradation curves of urban road networks under various scenarios, including "random attacks," "targeted attacks," and "community failures." This enriches the methodology of urban network resilience research from the perspective of the space of flows.

2.Practical Value: Decision Support for Tianhe District

As the economic core and transportation hub of Guangzhou, the robustness of Tianhe District's road network structure is directly related to regional safety and development. This study aims to:

a.Identify Implicit Critical Nodes: Detect intersections that may not be of the highest hierarchy but play a key "bridging" role in the network topology, avoiding single points of failure due to oversight.

b. Provide Quantitative Decision Support: Provide a scientific basis for urban managers to formulate disaster mitigation strategies, optimize road hierarchy, and plan emergency evacuation routes under limited resources, thereby enhancing the robustness of the urban system against unknown risks.

1.3 Study Area

Location Overview The study area is Tianhe District, Guangzhou, China. Located on the new urban central axis of Guangzhou, bordering the Pearl River to the south and Baiyun Mountain to the north, it serves as the new city center.

Network Characteristics As a typical central urban area of a mega-city, Tianhe District possesses an extremely dense road network featuring a complex hierarchical structure covering expressways, arterial roads, secondary roads, and branch roads. Its high-intensity population agglomeration and economic activities generate immense traffic flow demands, making the regional road network face significant operational pressure and complex topological characteristics under disruption. It serves as an ideal case for urban network resilience stress testing.

1.4 Research Objectives

1.Topological Model Construction: Build a graph-theoretical model *G(N, L)* of the Tianhe District road network based on GIS data, realizing spatial digital reconstruction with geometric topology correction.

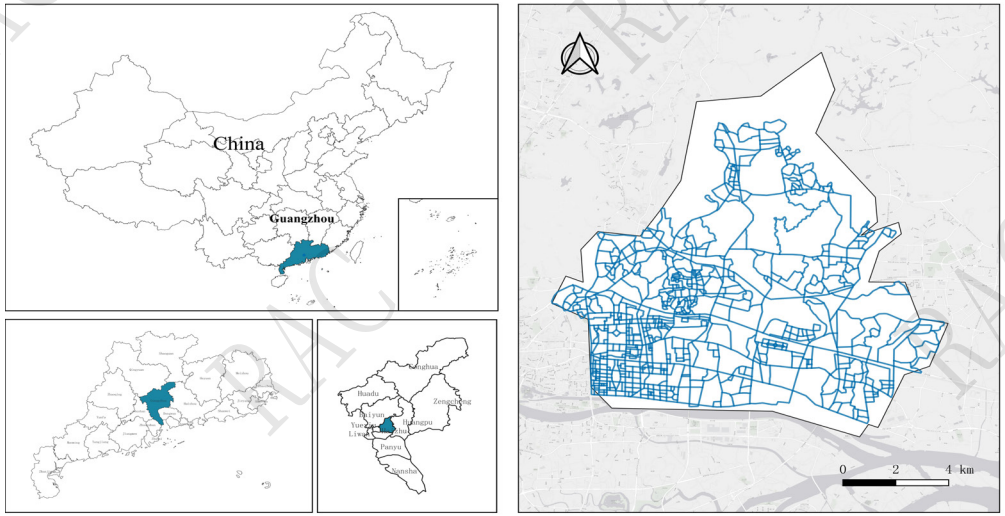


Fig.1 *Research Location Map*

2.Baseline Resilience Assessment: Calculate baseline performance metrics of the road network under normal conditions, including node Betweenness Centrality distribution, global Giant Connected Component (*GCC*), and Network Efficiency, to identify current structural characteristics.

3.Multi-Scenario Stress Testing Simulation: Design three stress testing scenarios—Random Failure, Targeted Attack (Hotspot-based), and Community Failure—to simulate the impact of different disaster types on the road network.

4.Spatial Resilience and Optimization Strategy: Identify "high-resilience zones" and "vulnerable nodes" within the network, quantify the functional loss of nodes before and after disasters ($Diff_x$), and propose targeted spatial optimization and resilience enhancement strategies.

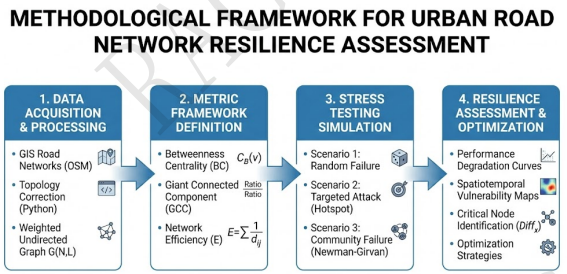


Fig.2 *Methodological framework*

1.5 Research Content

Part 1 Data Acquisition and Topology Repair Extract Tianhe District road centerlines based on OSM and related geographic information data. Use Python scripts for automated topological checks .

Part 2 Construction of Graph Theory Index System Select Betweenness Centrality as the core metric for measuring node importance, reflecting the "hub" status of nodes in the shortest paths of the network; select Giant Connected Component (*GCC*) ratio and Network Efficiency as macroscopic metrics for measuring the overall robustness and transmission performance of the network.

Part 3 Simulation of Disruptions Based on Stress Testing.

- ***Scenario 1 (Random Failure)*** Simulate random intersection failures caused by earthquakes or accidents to assess the network's overall anti-interference ability.
- ***Scenario 2 (Hotspot Failure)*** Combine population raster data to identify high-density areas (high-risk zones) and simulate a non-random destruction process where disasters prioritize densely populated areas.
- ***Scenario 3 (Community Failure)*** Apply the Newman-Girvan algorithm to partition network communities and simulate the local outbreak of disasters within specific functional communities and their global impact.

Part 4 Spatiotemporal Resilience Analysis and Critical Node Identification

Generate spatial maps of road network resilience and vulnerability by calculating the difference in node importance before and after disruptions ($Diff_x$), accurately identifying critical intersections with sharp functional decline post-disaster to provide targets for planning interventions.

2. Methodology

2.1 Data Acquisition

The spatial database for this study was constructed using multi-source geospatial big data to ensure the accuracy of the topological representation and the validity of the stress testing.

- **Road Network Data:** Sourced from *OpenStreetMap (OSM)*. The raw data was cleaned to extract road centerlines, serving as the geometric skeleton of the graph.
- **Population Data:** Acquired from *WorldPop (100m resolution)*. This dataset provides high-resolution population distribution weights for identifying "Hotspot" nodes.
- **Administrative Boundaries:** Derived from *Tianditu (Map World)* to strictly define the study area of Tianhe District.
- **Street View Imagery:** Sampled from *Baidu Map* to visually validate the physical environment of critical nodes identified by the model.

2.2 List of Symbols

Symbol	Definition	Symbol	Definition
G	*The urban road network graph (N, E)*	GCC	*Giant Connected Component ratio*
N	*Set of nodes (Intersections)*	E	*Global Network Efficiency*
L	*Set of links (Road segments)*	$C_B(v)$	*Betweenness Centrality of node v*
N_{total}	*Total number of nodes in the network*	$Diff_x$	*Difference in node importance*
d_{ij}	*Shortest path distance between i and j*	P	*Probability of node removal*

2.3 Methodology: Graph Theory & Stress Testing

A. Network Modeling Approach

We adopted a primal graph approach to model the spatial continuity of the urban fabric.

- Topological Representation: The road network is simplified into an undirected weighted graph *G(N, L)*, where road intersections are represented as nodes *(N)* and road segments are represented as edges *(L)*.
- Weighting: Edges are weighted by physical length to reflect actual travel distances.

B. Key Metrics for Resilience Assessment

To quantify resilience from both local and global perspectives, three key metrics were selected:

Betweenness Centrality (C_B)

Formula:

$$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

(where σ_{st} is the total number of shortest paths from s to t).

- **Interpretation:** Measures the frequency a node acts as a "bridge" along the shortest paths between two other nodes.
- **Justification:** It identifies "Hub" nodes that control the flow of the network. A drop in C_B indicates a loss of structural control.

Giant Connected Component (GCC)

Formula:

$$GCC = \frac{N_{LCC}}{N_{Total}}$$

- **Interpretation:** The ratio of nodes in the largest connected cluster to the total nodes.
- **Justification:** A macro-indicator of Network Robustness. It determines if the network remains a single functional entity or fragments into isolated islands under stress.

Network Efficiency (E)

Formula:

$$E = \frac{2}{N(N-1)} \sum_{i \neq j} \frac{1}{d_{ij}}$$

- **Interpretation:** The average inverse shortest path distance, reflecting the ease of information or flow transmission.
- **Justification:**It evaluates the Functional Performance of the network. Unlike *GCC* (which is binary: connected or not), Efficiency captures the degradation of travel quality (detours).

C. Disruption Scenarios & Simulation Mechanism

The "Stress Testing" involves removing nodes based on specific logic to simulate different disaster types. To ensure statistical significance, **Monte Carlo simulations** *(N=50)* were performed for each scenario to mitigate stochastic variations.

Scenario 1: Random Failure

- **Explanation:** Nodes are removed randomly from the network.
- **Context:** Simulates arbitrary, non-targeted failures such as random accidents or scattered structural failures during an earthquake.

Scenario 2: Hotspot (Targeted) Attack

- **Explanation:**Nodes are ranked by population density (derived from *WorldPop*), and high-density nodes are prioritized for removal.
- **Context:** Simulates disasters that disproportionately affect densely populated centers, testing the system's ability to protect core human settlements.

Scenario 3: Community Failure

- **Community Detection:** We utilized the Louvain Algorithm, a heuristic method based on modularity maximization, to detect community structures within the road network.
- **Mechanism:** The network is partitioned into modular communities (sub-regions with high internal density). Disruptions are simulated by removing nodes within specific communities to test localized modular breakdown.

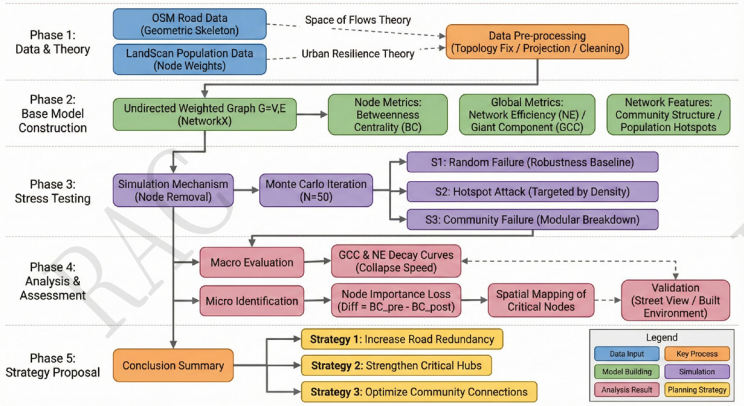


Fig.3 *Methodological Framework of the Research*

This flowchart illustrates the integrated five-phase workflow for assessing road network resilience. **Phase 1** involves the acquisition and pre-processing of multi-source data, including *OSM* road networks and WorldPop population data, grounded in the "Space of Flows" theory. **Phase 2** constructs the undirected weighted graph model and defines key topological metrics *(BC, GCC, NE)*. **Phase 3** executes the stress testing via Monte Carlo simulations *(N=50)* across three distinct scenarios: Random Failure, Hotspot Attack, and Community Failure (based on the Louvain algorithm). **Phase 4** focuses on the quantitative analysis of macro-performance degradation curves and micro-identification of node importance loss *(Diff)*. Finally, **Phase 5** proposes targeted spatial optimization strategies based on the simulation results to enhance urban resilience.

3. Analysis

3.1 Topological Reconstruction and Spatial Heterogeneity

Fig 4: Road Network Topology (Current Status)

Spatial Pattern: The network exhibits significant spatial heterogeneity. The southwestern region (CBD area) presents a high-density grid pattern, reflecting intensive urbanization. In contrast, the northeastern region shows a sparse, tree-like structure due to topographical constraints (mountainous terrain).

Fig 5: Spatial Distribution of Network Nodes

This map visualizes the discrete intersections *(Nodes, N)* extracted from the road network.

Node Density: The distribution of nodes serves as a proxy for intersection density. The high concentration of nodes in the southwest facilitates high redundancy and alternative routing options, whereas the sparse distribution in the north indicates potential structural vulnerability where alternative paths are limited.



Fig.4 Road Network Topology
Topological representation of the urban road network (G(N, L)) within the study area1.

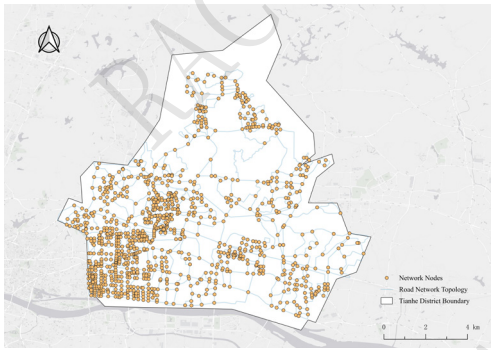


Fig. 5 Spatial Distribution of Network Nodes
Spatial distribution of topological nodes (intersections) extracted for the graph-based model.

Comparative Analysis And Structural Insight

The comparative visualization of the road skeleton *(Fig. 4)* and node dispersion *(Fig. 5)* exposes a critical Topological Dichotomy driven by urbanization and topography.

1.The "Lattice" vs. The "Tree": The southern CBD exhibits a high clustering coefficient with short average path lengths, typical of a "Small-World" network. This lattice-like structure inherently possesses high redundancy (alternative paths), suggesting high resilience against random failures.

2.The Vulnerability of Sparsity: Conversely, the northern region, constrained by mountainous terrain, displays a "Tree-like" topology with low degree centrality. *Fig. 2* confirms that node density drops strictly along the urban-rural gradient. This structural scarcity implies that the northern network lacks "loops," making it highly susceptible to fragmentation—a single cut on a main artery could isolate entire communities.

3.2 Integration of Demographic Attributes

Methodological Synthesis & Risk Identification

The superposition of demographic data onto the physical topology *(Fig. 5 → Fig. 7)* bridges the gap between "Network Centrality" (where the road is) and "Place Centrality" (where the people are).

1.From Geometry to Semantics: While traditional graph theory treats all nodes equally, our mapping creates a "Semantic Graph." *Fig. 4* highlights that while the geometric center of the district is geographically central, the "Center of Gravity" for population is heavily skewed to the southwest.

2.Identification of High-Consequence Nodes: This fusion allows for the identification of "High-Consequence Nodes." A node in the red zone of *Fig. 4* is not just a traffic intersection; it is a critical service point. Failure here does not just mean a detour; it implies a significant loss of access for a large population. This justifies the necessity of the "Hotspot Attack" scenario in the subsequent stress test, distinguishing our approach from purely topological resilience studies.

Demographic Hotspots: The map reveals a distinct monocentric aggregation, with the highest population density concentrated in the Zhujiang New Town and Tianhe Road commercial circles. This data provides the foundational "weight" for the Scenario 2 (Hotspot Attack) simulation.

Node Weights: The size and color of the nodes in Figure 4 correspond to the assigned population density. This transformation converts the continuous demographic surface into discrete nodal attributes, allowing us to quantify the "social cost" of specific node failures. Nodes with high population weights are identified as High-Risk Targets in the targeted attack model.

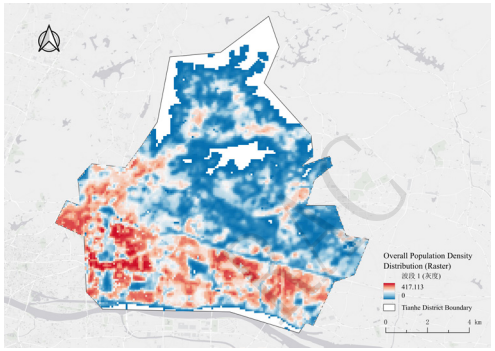


Fig.6 ***Overall Population Density Distribution (Raster)***
Raster-based population density estimation visualized using a color gradient to identify core aggregation zones.

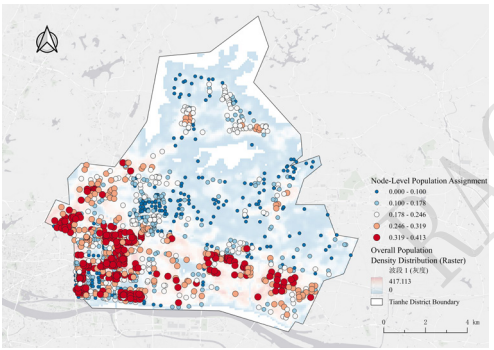


Fig.7 Node-Level Population Assignment
Discretization of population weights to network nodes to evaluate node importance relative to demographic demand.

3.3 Integration of Demographic Attributes

Network Community Structure Partitioning Using the Louvain Algorithm (based on modularity maximization), the road network was partitioned into distinct communities.

The map identifies approximately 20 functional communities (indicated by different colors). Nodes within the same color group exhibit stronger internal connectivity than external connectivity.

This structure reveals the "modular" nature of the urban grid. The boundaries between communities often correspond to major arterial roads or physical barriers (rivers, highways). In Scenario 3 (Community Failure), these color-coded clusters serve as the units for localized disruption simulation, testing the resilience of specific functional zones.

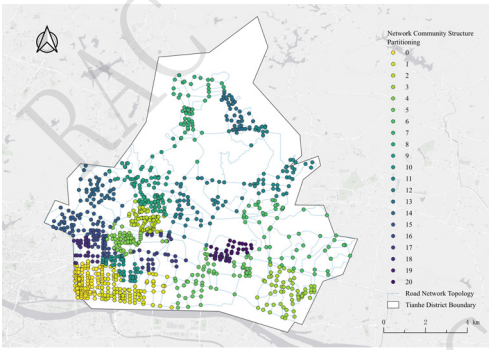


Fig.8 Network Community Structure Partitioning
Identification of community structures (sub-regional clusters) utilizing the modularity maximization algorithm.

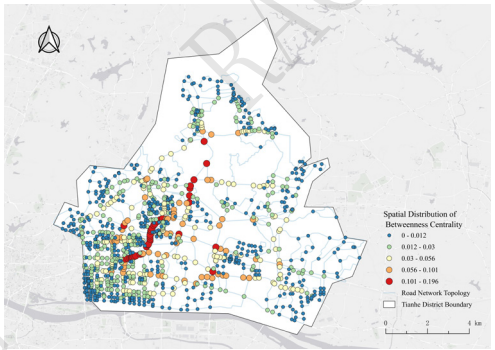


Fig.9 Spatial Distribution of Betweenness Centrality
Spatial distribution of Betweenness Centrality (C_B) under baseline conditions, visualized by the "natural breaks" method to highlight critical network components4.

This map visualizes the *Betweenness Centrality (BC)* of all nodes in the baseline network (before any disruption).C_B measures the frequency with which a node acts as a bridge along the shortest path between two other nodes.

The red nodes *(High C_B)* form the "structural spine" of Tianhe District. Unlike node density which is highest in the CBD, high-BC nodes are often located at key arterial intersections and bridges (e.g., across the Pearl River or highways).

These nodes control the global flow of the network. Their high centrality implies that their failure would trigger a significant drop in *Network Efficiency (NE)*. This distribution serves as the reference point for calculating the Importance Loss *($Diff_x$)* in the subsequent analysis phase.

3.4 Multi-Scenario Stress Testing and Resilience Evaluation

The spatial dimension of resilience is evaluated by quantifying the "Loss of Importance" *($Diff_x$)* for each node across three distinct disruption scenarios. This metric, derived from the difference in Betweenness Centrality *(C_B)* before and after disruption, reveals the dynamic redistribution of network flows under stress.

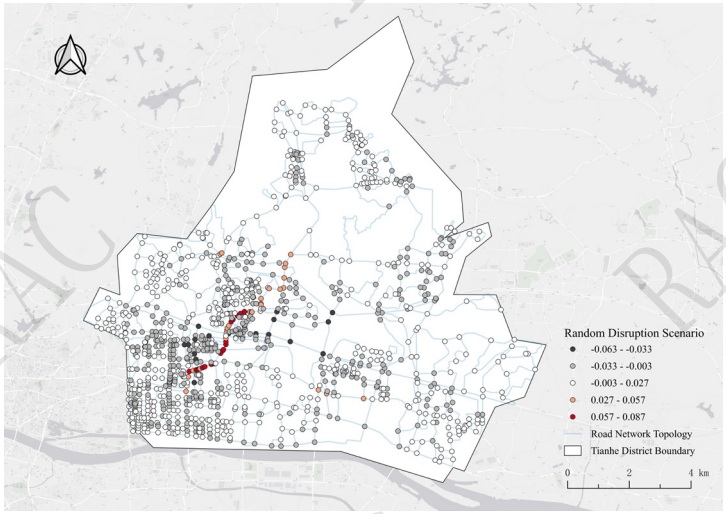


Fig.10 Random Disruption Scenario
Difference map ($Diff_x$) illustrating the redistribution of centrality under the Random Disruption scenario. Red nodes indicate a significant increase in load (stress transfer).

Under the stochastic failure model (simulating non-targeted events like random accidents or scattered structural failures), the network exhibits a dispersed pattern of vulnerability. The red nodes, representing high $Diff_x$ values, are scattered throughout the Tianhe District but show a tendency to cluster near key topological bridges rather than purely in the geometric center. This indicates that under random attacks, the "structural bridges"—nodes that connect different network modules—are statistically more prone to losing their bridging function, forcing flows to detour significantly.

The widespread distribution of grey dots (negligible change) suggests that a large portion of the network possesses a degree of local redundancy, absorbing minor random shocks without catastrophic functional loss.

In thisTargeted Disruption scenario, nodes were removed based on population density rankings, simulating disasters that disproportionately affect crowd-gathering areas (e.g., commercial centers). Interestingly, the spatial distribution of criticality loss is highly concentrated in the high-density CBD areas (southwestern Tianhe). However, the intensity of the "red" nodes is visibly less aggressive compared to the Random scenario in certain peripheral areas. This reveals a critical mismatch: Demographic Hotspots neq$ Topological Keystones. While high-population nodes are socially critical, they often reside in grid-like structures with high redundancy (multiple alternative paths).

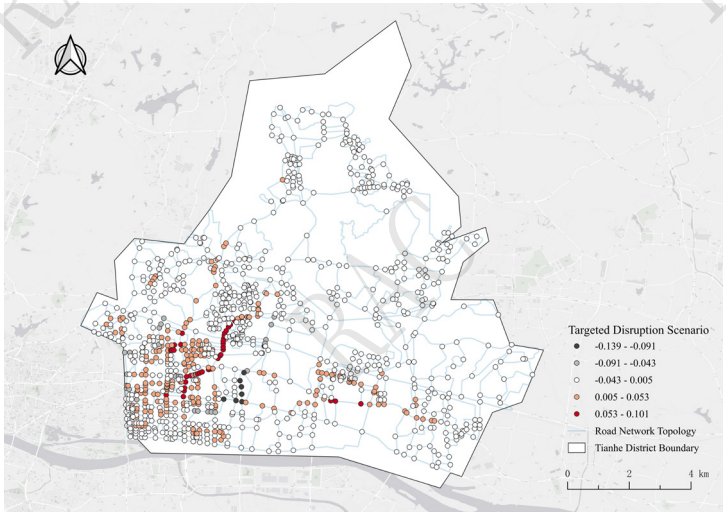


Fig.11 Targeted Disruption Scenario
Difference map ($Diff_x$) for the Targeted Disruption scenario (removal of high-C_B nodes), showing extensive structural impact and secondary bottlenecks.

Therefore, their removal results in localized functional loss but does not necessarily shatter the global connectivity of the network as severely as the loss of strategic bridge nodes.

In the CommunityBased Disruption Scenario, by attacking the network based on its modular community structure (Louvain algorithm), we observe a distinct "boundary effect." The nodes with the highest importance loss are often located at the interfaces between communities. When a community is compromised, the inter-community connectors (bridge nodes) lose their function as conduits for global flow. This scenario highlights the vulnerability of the "arterial skeleton" that binds the urban patchwork together. The visualization confirms that the network's resilience is not uniform; specific functional zones act as critical failure points that can trigger cascading effects across the entire district.

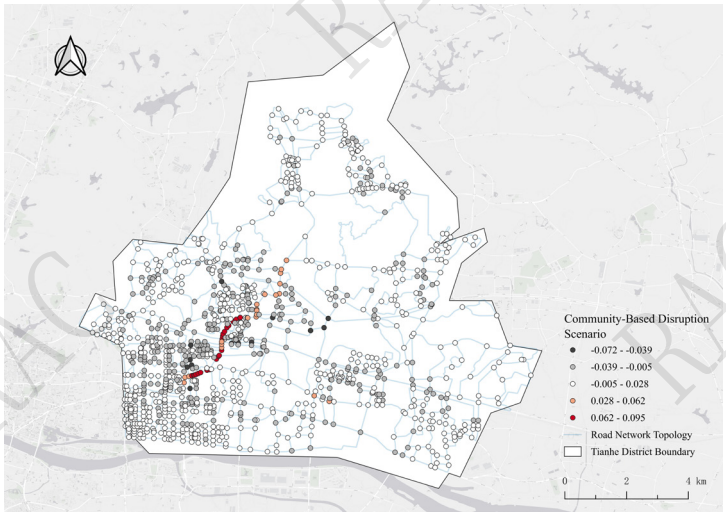


Fig.12 Community-Based Disruption Scenario
Difference map ($Diff_x$) for the Community-Based Disruption scenario, evaluating the cascading effects of failure within specific topological clusters.

The visualization confirms that the network's resilience is not uniform; specific functional zones act as critical failure points that can trigger cascading effects across the entire district.

Spatial Mechanics of Disruption Scenarios

The "Density-Redundancy" Paradox

Comparative analysis of the Targeted (Hotspot) and Random scenarios reveals a counter-intuitive finding: Demographic centrality does not equate to topological fragility. While the Targeted scenario aggressively attacks the CBD (Zhujiang New Town), the network retains significant functionality. This confirms the "Redundancy Hypothesis": high-density urban cores often possess grid-like topologies with high clustering coefficients, providing numerous alternative loops that "absorb" the shock of node removal. Conversely, the dispersed damage in the Random scenario often strikes the "sparse" northern regions or critical bridges, where lack of redundancy turns a single failure into a systemic disconnection.

The Vulnerability of Structural Interfaces

The Community-Based scenario exposes the "Boundary Effect" of the urban fabric. Unlike the localized impact of Hotspot attacks, community-based disruptions target the interfaces between functional modules. The high criticality loss ($Diff_x$) observed at these boundaries suggests that inter-community connectors (bridges/arterials) act as the "valves" of the system. Their failure isolates entire functional zones, proving that the network's resilience is constrained not by the strength of its modules, but by the robustness of the links that bind them together.

3.5 Robustness Analysis (GCC Degradation Curves).

To quantitatively assess the network's macroscopic behavior, we tracked the degradation of Robustness *(GCC)* and Efficiency *(NE)* against an increasing Node Removal Ratio (p).

The resulting curves *(Fig. 13&14)* uncover the non-linear collapse mechanisms of the urban road network.The *Giant Connected Component (GCC)* ratio serves as a proxy for the network's structural integrity. The degradation curves reveal a startling counter-intuitive phenomenon. The Random Failure (Blue Line) and Community Failure (Green Triangle) scenarios induce a rapid structural collapse. As the removal ratio p approaches 0.3-0.4, the *GCC* plummets from 1.0 to below 0.2, indicating a "Phase Transition" or percolation threshold where the network fragments into disconnected islands. This steep decline suggests that the Tianhe road network acts like a "scalefree network" which is typically robust to random failure only if the hub nodes are protected;however, due to the specific topology here, random attacks likely hit the numerous "bridge" nodes that maintain global connectivity.

Conversely, the Hotspot Attack (Red Square) curve decays much more gradually. Even when 40% of the highpopulation nodes are removed, the *GCC* remains above 0.4. This empirically proves the "Redundancy Hypothesis" of the CBD area: the dense grid structure in high-population zones offers numerous alternative paths, preserving global connectivity even when major commercial nodes are disabled.

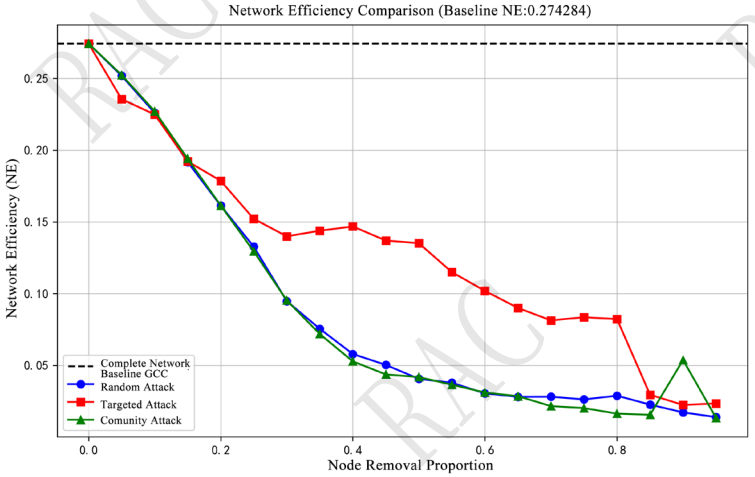


Fig.13 Efficiency_With_Baseline

3.6 Efficiency Analysis (NE Degradation Curves).

Network Efficiency (NE) measures the ease of information transmission (average inverse shortest path distance). The *NE* curves mirror the robustness trends but with greater sensitivity to initial disruptions. In the early stages ($p < 0.2$), the Hotspot Attack (Red Square) causes a linear and manageable decline in efficiency, reinforcing the finding that high-density nodes are topologically substitutable. However, the Random (Blue) and Community (Green) attacks cause an exponential decay in efficiency almost immediately. This implies that these attacks rapidly sever the "shortcuts" in the network, forcing remaining traffic onto significantly longer, tortuous paths.

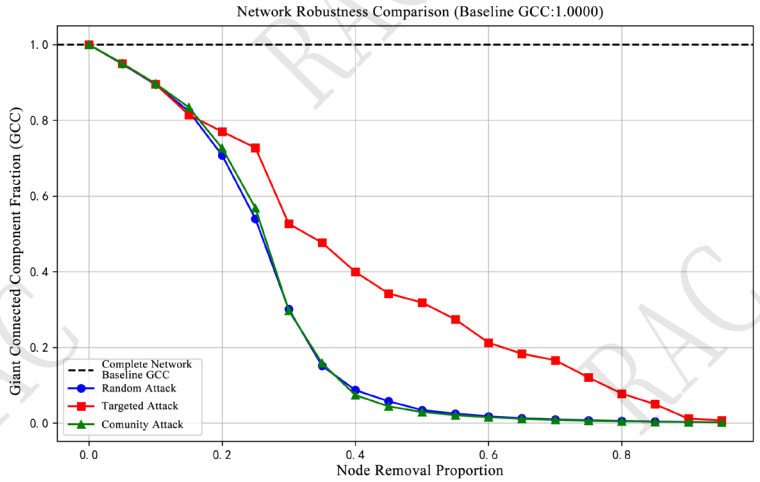


Fig.14 Robustness_With_Baseline

The divergence between the Red curve and the Blue/Green curves highlights a critical insight for urban resilience planning: protecting the topological skeleton (bridges, tunnels, arterial connectors) is more critical for maintaining global system efficiency than protecting the demographic centers alone. The resilience of the network depends less on the density of the nodes and more on their role in maintaining the "Small-World" property of the graph.

3.7 Dynamics of Performance Degradation

Critical Transitions and Percolation Thresholds

The decay curves demonstrate distinct non-linear dynamics. The Random and Community scenarios exhibit a sharp "Phase Transition" (percolation threshold) at $p \approx 0.3$-0.4, where the network state flips from "connected" to "fragmented". This steep slope signifies a lack of global elasticity against structural attacks. In contrast, the linear, gradual decline of the Hotspot curve indicates a "Graceful Degradation" mode. This divergence highlights that the Tianhe road network behaves as a Scale-Free Network, which is robust against attacks on "Hubs" (if hubs are defined by population) but fragile against attacks on "Bridges" (random or community connectors).

Efficiency as a Sensitive Precursor

Comparing the two metrics, *Network Efficiency (NE)* proves to be a more sensitive diagnostic tool than *Robustness (GCC)*. While *GCC* remains relatively stable in the early stages of the Hotspot attack (indicating physical connectivity persists), *NE* drops immediately. This implies that while the "skeleton" remains intact, the "quality of flow" deteriorates rapidly due to forced detours. This distinction is crucial for resilience planning: preserving connectivity *(GCC)* prevents isolation, but preserving efficiency *(NE)* is vital for maintaining emergency response speeds and economic continuity.

4. Verification and Evaluation

4.1 Ground-Truthing: Street View Verification of Topological Attributes

Comparative Analysis of Built Environments

Group A: Systemically Critical Nodes (The Resilience Skeleton)

Built Environment Characteristics: High-grade "Hard Infrastructure." These scenes are dominated by multi-lane carriageways, complex grade-separated interchanges (flyovers), and cross-river bridges. The spatial scale is vast, prioritizing vehicular throughput over pedestrian scale.

Road Hierarchy & Location: Predominantly Urban Expressways and Primary Arterials located at the geometric centroid of flows or key topographical choke points (e.g., bridgeheads).

Land Use Context: Transport-dominated space interspersed with high-density CBD commercial functions.

Summary: These nodes represent the "Valves of the City." Their physical rigidity makes distinct expansion difficult, yet they carry the highest topological load.


C-1


C-2


C-3


C-4

C-1 Tianhe Computer Center
C-2 Shangde Building
C-3 The Third Affiliated Hospital of Sun Yat-sen University
C-4 Haili Cultural and Sports Center, South China University of Technology



V-1 Tangxia
V-2 UTOPA Shopping Center
V-3 Huajing Badminton Center
V-4 Fenghuang Xincun


V-1


V-2


V-3


V-4

Group B: Structurally Vulnerable Nodes (The Peripheral Capillaries)

Built Environment Characteristics: Low-grade "Capillary Texture." Images typically depict narrow two-lane streets, dead-ends (cul-de-sacs), or roads abrupt ending at physical barriers (walls, hills).

Road Hierarchy & Location: Mostly Secondary Roads or Branch Roads located at the administrative boundaries, urban-rural fringes, or deep within enclosed communities.

Land Use Context: Old residential quarters, enclosed institutional compounds (campuses), or ecological conservation interfaces.

Summary: These nodes represent the "Blind Spots of Connectivity." Their vulnerability stems from a lack of loops and alternative paths, making them prone to isolation.

4.2 Spatiotemporal Identification of Criticality & Vulnerability

Most Important Nodes (Systemically Critical): Defined as the intersection of the Top 2 Classes (Levels 4 & 5) across all three scenarios. These nodes consistently suffer high functionality loss regardless of the disruption type, indicating their irreplaceable role in the global topology.

Vulnerable Nodes (Structurally Weak): Defined as the intersection of the Bottom 2 Classes (Levels 1 & 2) across all three scenarios.

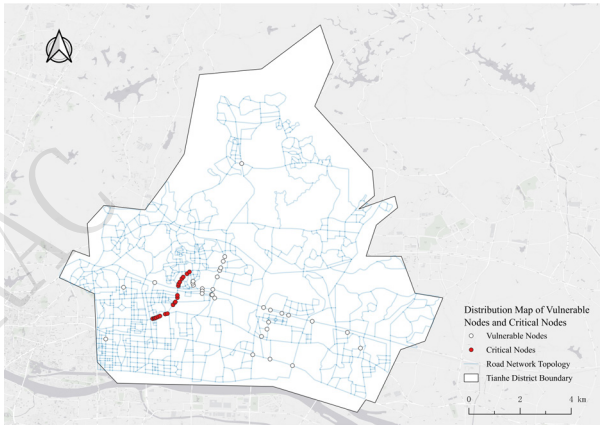


Fig.15 Spatiotemporal Identification of Systemically Critical and Structurally Vulnerable Nodes Based on Multi-Scenario Intersection Analysis.

This figure illustrates the distribution of two categories of nodes with extreme topological attributes within the Tianhe District road network. Red nodes (Critical Nodes) represent the intersection of the top 20% (Top 1 Quintile) of Betweenness Centrality Loss *($Diff_x$)* values across random, hotspot, and community disruption scenarios, constituting the "core skeleton" that maintains global connectivity. White nodes (Vulnerable Nodes) represent the intersection of the bottom 40% (Bottom 2 Quintiles) across all three scenarios, denoting the "peripheral extremities" that are insensitive to interference but prone to isolation. This classification reveals the spatial heterogeneity of the road network under uncertain shocks.

4.3 Summary And Indepth Analysis

1. Corridor Effect of the Resilience Skeleton

a. Spatial Characteristics

As indicated by the red nodes, "Systemically Critical Nodes" are not uniformly distributed throughout the high-density built-up area (e.g., within the Zhujiang New Town CBD). Instead, they exhibit a distinct linear clustering, forming a north-south "Resilience Corridor." These nodes are primarily distributed along key arterial roads in central Tianhe District (e.g., the intersection of Guangzhou Avenue and Tianhe North Road).

b. Mechanism Exploration

This finding challenges the applicability of traditional "Central Place Theory" from a resilience perspective. Although the CBD has the highest population density, its dense grid topology provides high Topological Redundancy, reducing the impact of single-node failures. In contrast, the "corridor" formed by the red nodes acts as a critical connector between the southern commercial core and northern residential clusters. Simulations show that failure of these nodes causes a sharp drop in Network Efficiency (NE), acting as "valves" that sever the north-south flow of material and information.

2. Mismatch between Density and Resilience

a. Comprehensive Assessment

Contrasting the population density map (P3) with this figure reveals that high population density areas do not equate to high resilience risk zones. Despite the extreme population density in the Zhujiang New Town core, relatively few nodes are identified as "absolutely critical" (red), indicating the structural robustness of its road network.

b. Planning Implications

True risk is concentrated at the "structural interfaces" connecting distinct functional zones. Future urban renewal and resilience enhancement strategies should not be limited to reinforcing internal CBD roads. Priority must be given to upgrading the disaster resistance standards of crossregional channels (red nodes) and introducing redundant interfaces for peripheral areas (white nodes) to prevent systemic paralysis and localized isolation during major disasters.

5. Conclusion and Recommendations

5.1 Research Conclusions

This study integrated graph theory with multi-scenario stress testing to quantify the resilience of the Tianhe District road network. The simulation results reveal three fundamental mechanisms governing urban structural stability:

The "Density-Vulnerability" Mismatch

Contrary to the assumption that high-density areas are the most fragile, our Hotspot Attack simulation demonstrated that the CBD (Zhujiang New Town) exhibits high Topological Redundancy. The dense grid structure provides numerous alternative paths, maintaining *GCC* above 0.4 even after 40% node removal. This indicates that demographic centrality does not equate to structural fragility; rather, the risk lies in the lack of substitutability.

The "Achilles' Heel" of Inter-Community Connectors

The Community Failure and Random Failure scenarios triggered a rapid phase transition (collapse) of the network. The *Betweenness Centrality (C_B)* analysis confirms that the "Bridge Nodes"—which connect distinct functional communities or span physical barriers (e.g., the Pearl River)—are the system's most critical failure points. Their removal fragments the network into isolated islands, causing a catastrophic drop in *Network Efficiency (E)*.

Spatial Heterogeneity of Resilience

The *$Diff_x$* metric identified a clear core-periphery dichotomy. The "Systemically Critical Nodes" function as the immutable skeleton of the city, while "Structurally Vulnerable Nodes" at the northern mountainous periphery and community boundaries suffer from Topological Isolation. Resilience is not a uniform attribute but a spatially heterogeneous variable dependent on local connectivity patterns.

5.2 Targeted Urban Design Strategies

Moving from diagnosis to prescription, we propose a "Three-Tier Resilience Enhancement Strategy" based on the identification of Critical and Vulnerable nodes.

Tier 1: Structural Hardening for "Critical Nodes" (Nodes with High C_B)

Target: The "Red Nodes" identified in the $Diff_x$ map (e.g., major arterial interchanges, cross-river bridges).

Strategy: "Redundancy via Multi-Modality." Since these nodes are spatially constrained and difficult to expand physically, resilience must be improved by diversifying flow types.

Action: Prioritize the construction of parallel underground infrastructure (metro tunnels) or elevated emergency corridors to decouple local traffic from through-traffic.

Retrofitting: Implement Grade-A seismic and flood-proofing standards for these specific intersections, as their failure has a systemic cascading cost.

Tier 2: "Acupuncture" Connectivity for "Vulnerable Areas" (Nodes with Low Redundancy)

Target: The "Blue Nodes" at community boundaries and peripheral zones that are easily isolated.

Strategy: "Sewing the Urban Fabric." These nodes fail due to a lack of alternative connections.

Action: Introduce "Micro-Connectors" (e.g., opening dead-end roads, creating community-connector streets) to increase the Degree Centrality of these peripheral zones.

Emergency Loops: Establish "Emergency loops" that link vulnerable peripheral communities directly to the nearest arterial road, bypassing the potentially congested internal network during disasters.

Tier 3: Modular Integration for Community Resilience

Target: The boundaries between the detected communities (Louvain partitions).

Strategy: "Elastic Boundaries." The simulation showed that fractures occur at community interfaces.

Action: Transform rigid administrative boundaries into "Resilient Interfaces." Design transitional buffer zones with dense, small-scale road grids rather than relying on a single large arterial road to separate communities.

Dynamic Traffic Management: Deploy adaptive traffic signal control at these boundary nodes to dynamically re-route flows when a specific community module is compromised, preventing the "locking" of a single failure point.

5.3 Future Outlook: Towards Adaptive Resilience

Resilience is not a static attribute but an evolving capability. While this study establishes a topological baseline based on static networks, future research must transcend these boundaries by integrating real-time traffic flows and mobile signaling data to capture the dynamic fluctuations of the "Space of Flows." Furthermore, by coupling Graph Neural Networks (GNNs) with the stress testing framework established here, we aim to transition from "post-disaster assessment" to "pre-disaster predictive functionality." The ultimate goal is to construct a "Resilience Digital Twin," a cognitive system that enables city managers to visualize cascading failure risks and simulate adaptive interventions in real-time, thereby ensuring the continuous vitality of the urban organism under uncertainty.

Future urban planning should transition from "Static Protection" (defending against known risks) to "Adaptive Management" (managing unknown stresses). By integrating real-time traffic data with the Graph-Based Stress Testing framework established in this study, city managers can establish a "Digital Twin for Resilience," enabling dynamic vulnerability monitoring and pre-disaster simulation to ensure the continuous functionality of the urban flow space.